

Algebraic Geometry 20

Riemann surfaces

Contents

1 Curvature Tensors: From Metric to Spacetime

The main computational and conceptual pipeline is

$$g \longrightarrow \Gamma \longrightarrow \text{geodesics} \longrightarrow \text{Riemann} \longrightarrow \text{Ricci} \longrightarrow \text{scalar curvature} \longrightarrow \text{Weyl}.$$

The metric g is the starting object. Everything else is computed from it.

1.1 Level 1: Metric Tensor

A metric tensor tells us how to measure lengths, angles, and distances. In local coordinates

$$x^1, \dots, x^n,$$

it is written as

$$ds^2 = g_{ij} dx^i dx^j.$$

For tangent vectors

$$X = X^i \partial_i, \quad Y = Y^j \partial_j,$$

the metric gives

$$g(X, Y) = g_{ij} X^i Y^j.$$

Thus the metric is a machine of the form

two vectors $\longrightarrow$ one number.

For example, if

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = (1, 2), \quad Y = (3, 4),$$

then

$$g(X, Y) = 1 \cdot 3 + 2 \cdot 4 = 11.$$

So the metric is the first object needed before curvature can be discussed.

1.2 Level 2: Christoffel Symbols

The Christoffel symbols describe how coordinate directions change from point to point. They are computed from the metric by

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}).$$

They are not tensors, but they are essential for computing geodesics and curvature.

Conceptually,

metric changes $\longrightarrow$ Christoffel symbols.

If the metric is constant, then all derivatives vanish, and hence

$$\Gamma_{ij}^k = 0.$$

This is what happens in ordinary Euclidean coordinates.

1.3 Level 3: Geodesics

Geodesics are the straightest possible paths. They satisfy

$$\frac{d^2 x^k}{dt^2} + \Gamma_{ij}^k \frac{dx^i}{dt} \frac{dx^j}{dt} = 0.$$

If

$$\Gamma_{ij}^k = 0,$$

then

$$\frac{d^2 x^k}{dt^2} = 0,$$

so geodesics are straight lines.

If Christoffel symbols are nonzero, geodesics bend. Thus

$$\Gamma \longrightarrow \text{geodesic equations.}$$

Geometry	Geodesics
Flat plane	straight lines
Sphere	great circles
Hyperbolic plane	curved geodesics diverging from each other
Spacetime	worldlines and light rays

1.4 Level 4: Riemann Curvature Tensor

The Riemann tensor measures true intrinsic curvature. It is computed from the Christoffel symbols:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.$$

It takes three vectors and returns a vector:

$$R(X, Y)Z.$$

Geometrically, it measures what happens when a vector is parallel transported around a tiny loop.

$$\text{three vectors} \longrightarrow \text{one vector.}$$

After lowering an index, the tensor

$$R_{ijkl}$$

takes four vectors and returns one number.

1.5 Level 5: Ricci Tensor

The Ricci tensor is obtained by taking a trace of the Riemann tensor:

$$Ric_{\sigma\nu} = R^\rho_{\sigma\rho\nu}.$$

So Ricci is a compressed version of Riemann. It takes two vectors and returns a number:

$$Ric(X, Y) = Ric_{ij}X^iY^j.$$

Thus

two vectors $\longrightarrow$ one number.

Geometrically, Ricci curvature measures averaged curvature around a direction.

Ricci sign	Geometric meaning
$Ric > 0$	nearby geodesics tend to focus
$Ric < 0$	nearby geodesics tend to diverge
$Ric = 0$	no Ricci volume focusing

A Ricci matrix is the table

$$Ric_{ij} = Ric(e_i, e_j).$$

The entries are not chosen arbitrarily; they are obtained by contracting the Riemann tensor.

1.6 Level 6: Scalar Curvature

The scalar curvature is the trace of the Ricci tensor:

$$R = g^{ij} Ric_{ij}.$$

It is one number at each point:

point $\longrightarrow$ one curvature number.

In two dimensions,

$$R = 2K,$$

where K is Gaussian curvature.

Geometry	Scalar curvature
sphere	positive
plane	0
hyperbolic plane	negative

1.7 Level 7: Weyl Tensor

The Weyl tensor is the trace-free part of the Riemann tensor. For $n \geq 3$,

$$C_{ijkl} = R_{ijkl} - \frac{1}{n-2} (g_{ik}Ric_{jl} - g_{il}Ric_{jk} - g_{jk}Ric_{il} + g_{jl}Ric_{ik}) + \frac{R}{(n-1)(n-2)} (g_{ik}g_{jl} - g_{il}g_{jk}).$$

Conceptually,

$$\boxed{Riemann = Weyl + Ricci \text{ part} + scalar \text{ part}.}$$

Ricci measures volume focusing, while Weyl measures shape distortion.

Curvature part	Effect on small ball
Ricci	small ball becomes smaller or larger
Weyl	small ball becomes an ellipsoid with nearly same volume

Important facts:

$$C = 0$$

in dimension 2, and also identically in dimension 3. Weyl curvature becomes genuinely important in dimension 4, especially in spacetime.

1.8 Level 8: Spacetime Examples

In spacetime, the metric has Lorentzian signature

$$(-, +, +, +).$$

Instead of ordinary distance, it measures spacetime interval:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

The same pipeline applies:

$$g_{\mu\nu} \longrightarrow \Gamma_{\mu\nu}^\rho \longrightarrow R_{\sigma\mu\nu}^\rho \longrightarrow Ric_{\mu\nu} \longrightarrow R \longrightarrow C_{\rho\sigma\mu\nu}.$$

In general relativity, Ricci curvature is tied to matter-energy through Einstein's equation, while Weyl curvature describes tidal and gravitational-wave type effects.

2 Concrete Model Examples

2.1 Example A: Sphere of Radius r

The sphere of radius r has metric

$$ds^2 = r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

Thus

$$g = \begin{pmatrix} r^2 & 0 \\ 0 & r^2 \sin^2 \theta \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} \frac{1}{r^2} & 0 \\ 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}.$$

The nonzero Christoffel symbols are

$$\Gamma_{\phi\phi}^{\theta} = -\sin \theta \cos \theta,$$

and

$$\Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \cot \theta.$$

The geodesic equations are

$$\theta'' - \sin \theta \cos \theta (\phi')^2 = 0,$$

and

$$\phi'' + 2 \cot \theta \theta' \phi' = 0.$$

These describe great circles.

The Gaussian curvature is

$$K = \frac{1}{r^2}.$$

In two dimensions,

$$Ric = Kg.$$

Therefore

$$Ric = \frac{1}{r^2}g.$$

So

$$Ric_{\theta\theta} = 1, \quad Ric_{\phi\phi} = \sin^2 \theta.$$

The scalar curvature is

$$R = 2K = \frac{2}{r^2}.$$

The Weyl tensor is

$$C = 0,$$

because in dimension 2, Weyl curvature is identically zero.

Hence

$K = \frac{1}{r^2}, \quad Ric = \frac{1}{r^2}g, \quad R = \frac{2}{r^2}, \quad C = 0.$
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For $r = 1$,

$$K = 1, \quad R = 2.$$

For $r = 2$,

$$K = \frac{1}{4}, \quad R = \frac{1}{2}.$$

A larger sphere is less curved.

Sphere radius $r \longrightarrow$ color by $K = \frac{1}{r^2} \longrightarrow$ great-circle geodesics $\longrightarrow$ parallel transport rotation.

2.2 Example B: Flat Plane

The Euclidean plane has metric

$$ds^2 = dx^2 + dy^2.$$

Therefore

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since g is constant,

$$\Gamma_{ij}^k = 0.$$

Therefore

$$R^i_{jkl} = 0.$$

Hence

$$Ric = 0, \quad R = 0, \quad C = 0.$$

So

$\Gamma = 0,$	$Riemann = 0,$	$Ric = 0,$	$R = 0.$
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Visualization:

straight geodesics $\longrightarrow$ no parallel transport rotation $\longrightarrow$ zero curvature color.

The plane is locally and globally flat.

2.3 Example C: Hyperbolic Plane

Use the upper half-plane model:

$$ds^2 = \frac{dx^2 + dy^2}{y^2}, \quad y > 0.$$

Then

$$g = \begin{pmatrix} \frac{1}{y^2} & 0 \\ 0 & \frac{1}{y^2} \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} y^2 & 0 \\ 0 & y^2 \end{pmatrix}.$$

The Gaussian curvature is

$$K = -1.$$

In dimension 2,

$$Ric = Kg.$$

So

$$Ric = -g.$$

Thus

$$Ric = \begin{pmatrix} -\frac{1}{y^2} & 0 \\ 0 & -\frac{1}{y^2} \end{pmatrix}.$$

The scalar curvature is

$$R = 2K = -2.$$

The Weyl tensor is again

$$C = 0,$$

because it is a two-dimensional geometry.

Hence

$$\boxed{K = -1, \quad Ric = -g, \quad R = -2, \quad C = 0.}$$

Visualization:

negative curvature $\longrightarrow$ geodesics diverge $\longrightarrow$ triangles have angle sum $< \pi \longrightarrow$ constant negative curvature

Hyperbolic geometry is the standard model of constant negative curvature.

2.4 Example D: Schwarzschild Spacetime

The Schwarzschild metric is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2.$$

Let

$$f(r) = 1 - \frac{2M}{r}.$$

Then

$$g_{\mu\nu} = \begin{pmatrix} -f & 0 & 0 & 0 \\ 0 & f^{-1} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}.$$

Outside the mass, Schwarzschild is a vacuum solution:

$$Ric_{\mu\nu} = 0.$$

Therefore

$$R = 0.$$

But the Riemann tensor is not zero:

$$R_{\rho\sigma\mu\nu} \neq 0.$$

Also,

$$C_{\rho\sigma\mu\nu} \neq 0.$$

In fact, in vacuum Schwarzschild spacetime,

$$C_{\rho\sigma\mu\nu} = R_{\rho\sigma\mu\nu}.$$

Thus the curvature is pure Weyl curvature.

Ricci curvature = 0 but tidal Weyl curvature $\neq 0$.

This is important:

there is no matter locally outside the black hole, but spacetime is still curved.

A useful scalar invariant is the Kretschmann scalar:

$$K_{\text{Kr}} = R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} = \frac{48M^2}{r^6}.$$

Hence

$Ric_{\mu\nu} = 0,$ $R = 0,$ $Riemann \neq 0,$ $Weyl \neq 0.$

Visualization:

black hole spacetime $\longrightarrow$ Ricci color = 0 $\longrightarrow$ Weyl/tidal intensity nonzero $\longrightarrow$ light cones tilt $\longrightarrow$ radial stretching

Near the black hole, a small freely falling sphere becomes stretched radially and squeezed tangentially.

2.5 Example E: FLRW Universe

The FLRW metric is

$$ds^2 = -dt^2 + a(t)^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right).$$

Here

$$a(t)$$

is the scale factor, and

$$k = 1, 0, -1$$

represents positive, flat, or negative spatial curvature.

An important Ricci component is

$$Ric_{00} = -3\frac{\ddot{a}}{a}.$$

The scalar curvature is

$$R = 6 \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2 + k}{a^2} \right).$$

Interpretation:

$a(t) \longrightarrow$ space expands or contracts $\longrightarrow$ Ricci curvature changes $\longrightarrow$ geodesic distances change.

Unlike Schwarzschild vacuum, FLRW generally has nonzero Ricci curvature:

$$\boxed{Ric \neq 0}$$

in general, because the universe contains matter, radiation, dark matter, and dark energy in cosmological models.

Visualization:

expanding grid $\longrightarrow$ scale factor $a(t) \longrightarrow$ Ricci scalar heatmap over time $\longrightarrow$ geodesic separation $\longrightarrow$ c

3 Comparison Table

Example	Geometry	K	Ric	R	
Flat plane	Euclidean surface	0	0	0	
Sphere S_r^2	positive curvature surface	$\frac{1}{r^2}$	$\frac{1}{r^2}g$	$\frac{2}{r^2}$	
Hyperbolic plane	negative curvature surface	-1	$-g$	-2	
Schwarzschild	vacuum spacetime	not surface K	0	0	
FLRW universe	cosmology	depends on $k, a(t)$	$\neq 0$ usually	$\neq 0$ usually	0 in ide

4 Visualization Levels for Math3D

4.1 Level 1: Metric Visualization

Show local rulers. At every point, the metric determines a small ellipse or ellipsoid.

circle in coordinate space $\longrightarrow$ metric transforms it $\longrightarrow$ ellipse in measured geometry.

For the sphere, the metric changes with θ :

$$g_{\phi\phi} = r^2 \sin^2 \theta.$$

Near the pole, ϕ -movement is small. Near the equator, ϕ -movement is larger.

4.2 Level 2: Christoffel Visualization

Show basis vectors changing over the surface.

coordinate grid $\longrightarrow$ basis vectors vary $\longrightarrow$ Christoffel symbols appear.

For the sphere, lines of longitude meet at the poles, so the coordinate frame changes. That produces nonzero Christoffel symbols.

4.3 Level 3: Geodesic Visualization

Show geodesic curves.

Geometry	Visualization
Plane	straight lines
Sphere	great circles
Hyperbolic plane	semicircles and vertical lines in upper half-plane
Schwarzschild	light bending and orbital paths

4.4 Level 4: Riemann Visualization

Show parallel transport around a small loop.

start with arrow $Z \longrightarrow$ move around tiny loop spanned by $X, Y \longrightarrow$ compare final arrow with initial arrow

The difference is curvature.

On a sphere, an arrow transported around a triangle rotates. On a plane, it does not rotate.

4.5 Level 5: Ricci Visualization

Show geodesic focusing.

Ricci curvature	Visualization
positive Ricci	geodesics converge
negative Ricci	geodesics diverge
zero Ricci	no Ricci focusing

For a sphere, geodesics tend to meet. For hyperbolic space, geodesics tend to separate. For Schwarzschild vacuum, Ricci is zero outside the mass, but tidal distortion still exists.

4.6 Level 6: Scalar Curvature Visualization

Color the surface or spacetime slice by scalar curvature.

Geometry	Scalar curvature
Sphere	$R = \frac{2}{r^2} > 0$
Plane	$R = 0$
Hyperbolic plane	$R = -2$
Schwarzschild vacuum	$R = 0$
FLRW	$R = 6 \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2 + k}{a^2} \right)$

4.7 Level 7: Weyl Visualization

Show tidal distortion.

small sphere of test particles $\longrightarrow$ Weyl curvature $\longrightarrow$ ellipsoid.

For Schwarzschild:

radial stretching and tangential compression.

This shows black-hole tidal forces. In two-dimensional surfaces, Weyl is zero, so this level becomes important mainly in spacetime.

4.8 Level 8: Spacetime Visualization

Display:

light cones, worldlines, null geodesics, timelike geodesics, tidal ellipsoids, curvature

For Schwarzschild,

$$K_{\text{Kr}} = \frac{48M^2}{r^6}.$$

So one can color space by tidal intensity.

For FLRW, the scale factor $a(t)$ controls the expansion of space.

5 Final Conceptual Summary

The hierarchy is

$$g_{ij}$$

which measures lengths,

$$\Gamma_{ij}^k$$

which describes how coordinates and basis vectors change,

$$\text{geodesics}$$

which are straightest paths determined by Γ ,

$$R^\rho_{\sigma\mu\nu}$$

which measures full curvature,

$$\text{Ric}_{\mu\nu}$$

which measures averaged curvature and geodesic focusing,

$$\boxed{R}$$

which is one scalar curvature number,

and

$$\boxed{C_{\rho\sigma\mu\nu}}$$

which measures trace-free tidal distortion.

For visualization:

Object	Picture
Metric	local ruler
Christoffel	changing frame
Geodesics	straightest paths
Riemann	loop rotation
Ricci	focusing/diverging
Scalar	curvature heatmap
Weyl	tidal ellipsoid
Spacetime	light cones and worldlines